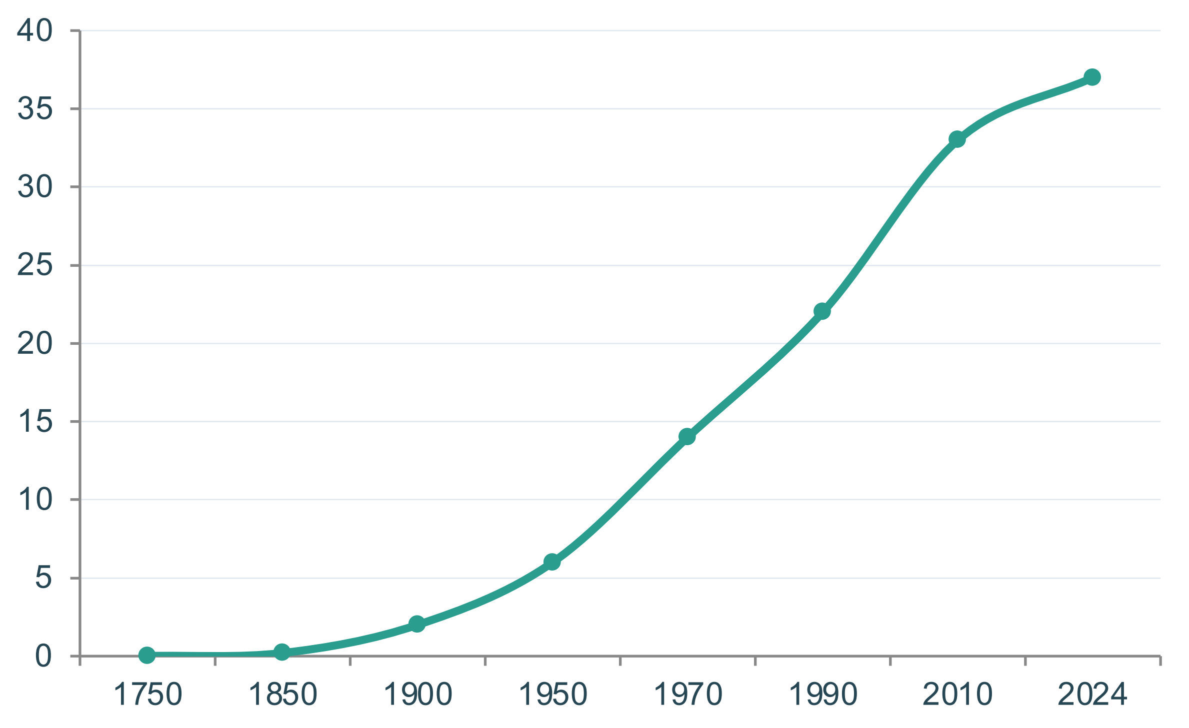
Global CO_2 Emissions: Who Emits, How Much, and What's Changing

Fossil fuels & industry · 1750–2024

Data: Global Carbon Budget (2025) via Our World in Data
Audience: Policy analysts, climate negotiators & sustainability researchers

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Long-run trajectory of global CO_2 emissions from fossil fuels and industry



1950: ~6 Bt

Post-war industrial acceleration begins

1990: ~22 Bt

Nearly quadrupled since 1950

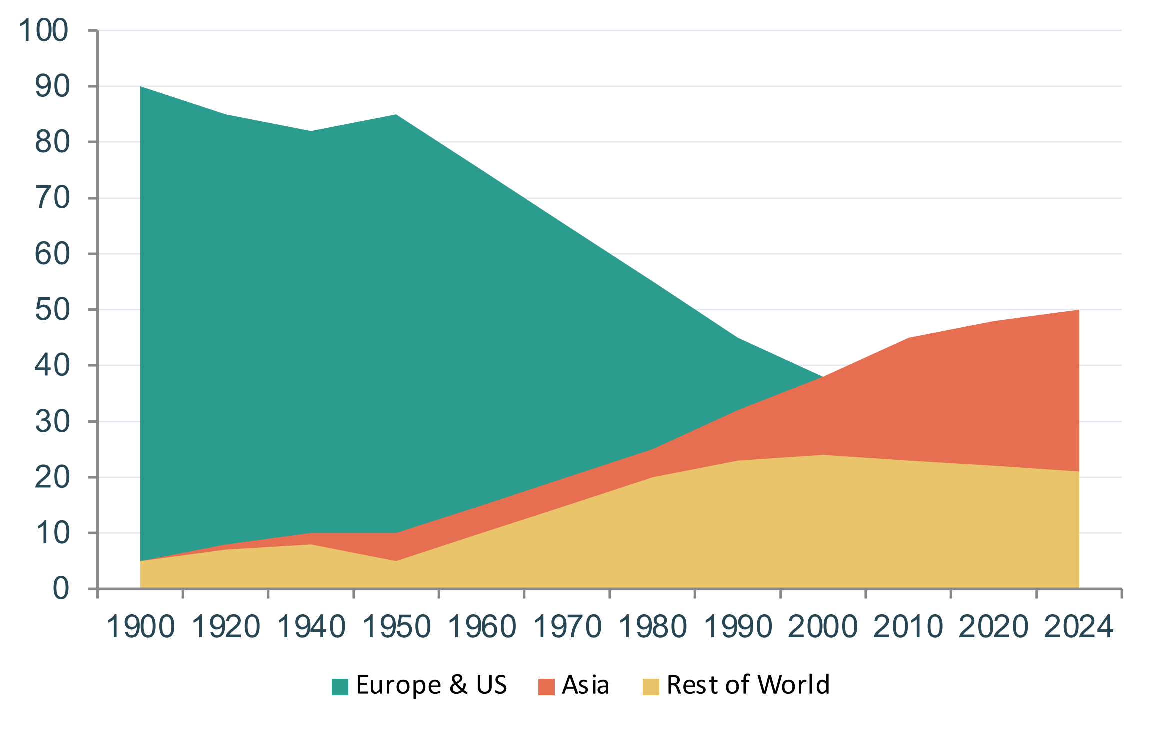
2024: ~37 Bt

Growth slowing — but no peak yet

Source: Global Carbon Budget (2025) via Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

Share of global CO_2 emissions by region over time



>90%

Europe + US share in 1900

<33%

Europe + US share today

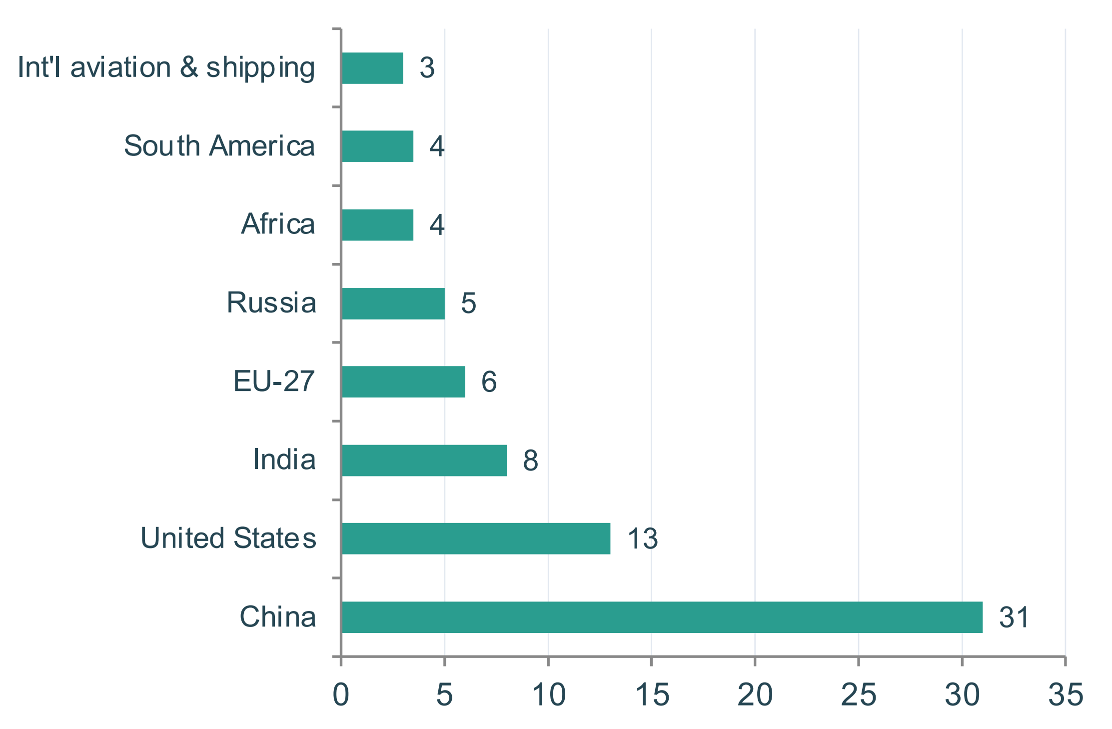
~50%

Asia's share today (led by China)

Source: Global Carbon Budget (2025) via Our World in Data

China Now Emits More Than One-Quarter of Global CO_2

Country-level shares of global CO_2 emissions, 2024



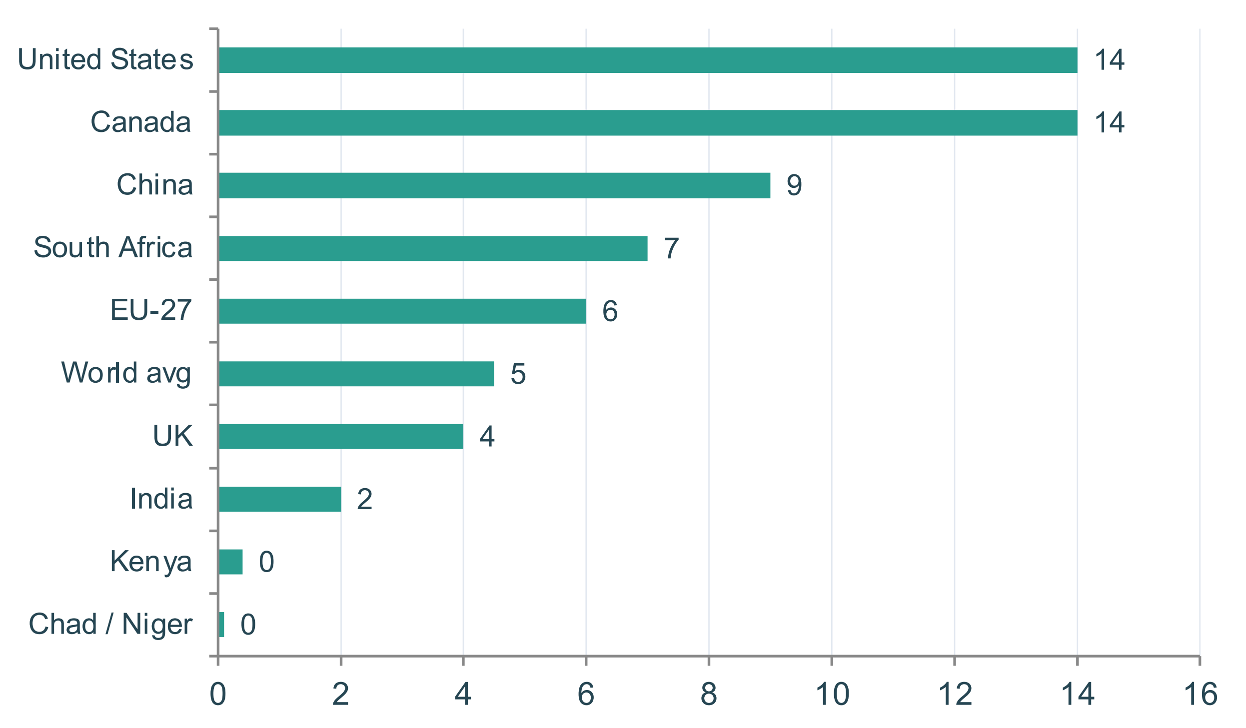
Country / Region	Share
China	~31%
United States	~13%
India	~8%
EU-27	~6%
Russia	~5%
Africa	~3–4%
South America	~3–4%
Int'l aviation & shipping	~3%

Africa and South America each account for only 3–4% — comparable to international aviation and shipping combined.

Source: Global Carbon Budget (2025) via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

Per capita CO_2 emissions, 2024 (tonnes per person)



150×

gap between lowest and highest large-country emitters

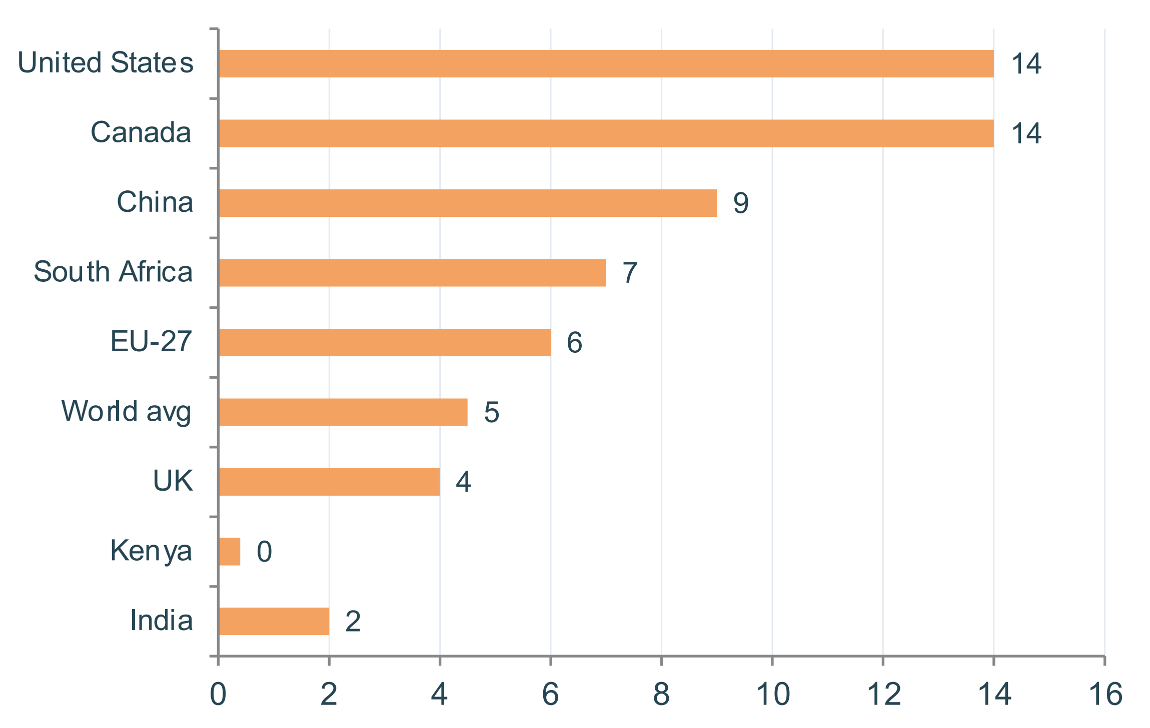
Context

The average American emits in under two days what the average person in Mali or Niger emits in an entire year.

Source: Global Carbon Budget (2025) via Our World in Data

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

Snapshot of per capita CO_2 emissions for key countries, 2024



China's per capita emissions remain below the US despite being the largest total emitter.

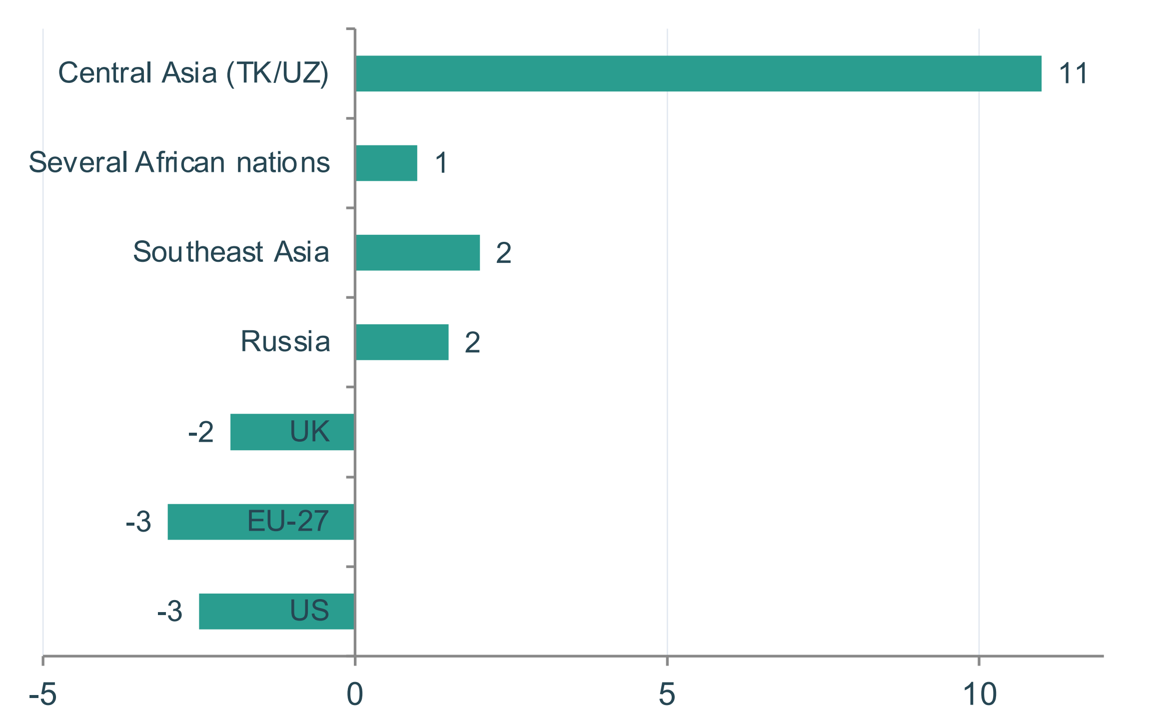
India remains very low at ~2 t per person — well below the global average of ~4.5 t.

France, UK, and Portugal emit far less per capita than Germany or the US, thanks to nuclear and renewables.

Source: Global Carbon Budget (2025) via Our World in Data

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Annual percentage change in CO_2 emissions, 2024



Key

- Decline (US, EU, UK)
- Increase (Russia, SE Asia, Africa)
- Large increase (>10%)

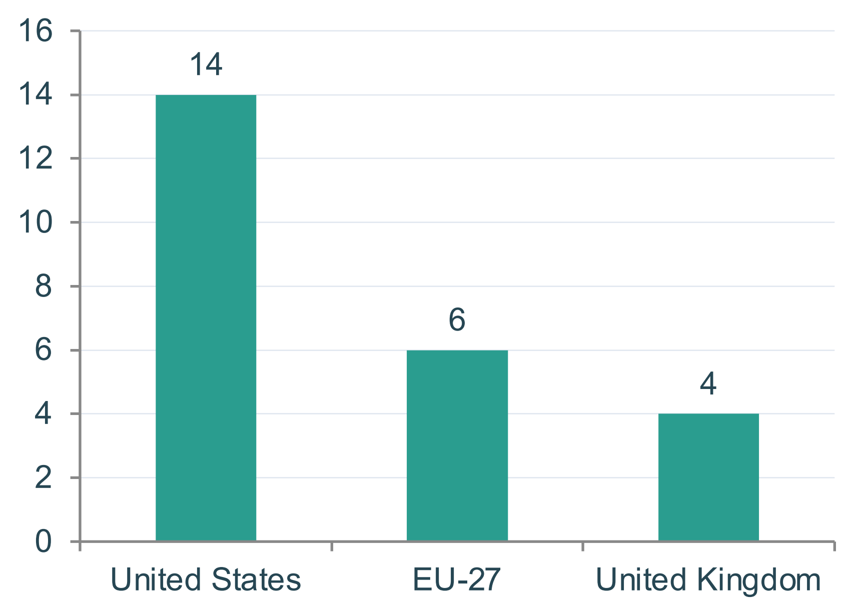
Notable trends

- US, much of EU, and UK saw emissions decline
- Russia, parts of SE Asia, and several African nations experienced increases
- Central Asia (Turkmenistan/Uzbekistan) saw particularly large increases exceeding 10%

Source: Global Carbon Budget (2025) via Our World in Data

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Among wealthy nations, energy policy choices drive large per-capita differences



Lower emitters

France, the UK, and Portugal have much lower per capita emissions than Germany, the Netherlands, or the US — largely because a higher share of their electricity comes from nuclear and renewable sources rather than fossil fuels.

The insight

Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix. This demonstrates that policy and technology choices can decouple prosperity from high emissions.

US ~14 t → EU-27 ~6 t → UK ~4 t | Similar living standards, very different carbon footprints

Source: Global Carbon Budget (2025) via Our World in Data

Key Takeaways: Responsibility Is Shared but Unequal

01

No peak yet

Global CO_2 emissions from fossil fuels exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth.

02

China is the largest annual emitter

China emits ~31% of the global total, but its per capita emissions (~9 t) remain well below the US (~14 t).

03

Historical responsibility is dominated by Europe and the US

Europe and the US accounted for >90% of emissions in 1900; today they represent less than one-third.

04

Per capita gaps of 150× persist

The gap between the lowest emitters (Sub-Saharan Africa, ~0.1 t/person) and the highest large-country emitters (US/Canada/Australia, ~14–15 t/person) remains enormous.

05

Energy policy and technology choices matter

Countries with similar prosperity levels can have very different per capita emissions depending on their energy mix — France and the UK emit far less per capita than Germany or the US.

Data: Global Carbon Budget (2025) via Our World in Data · Authors: Hannah Ritchie & Max Roser